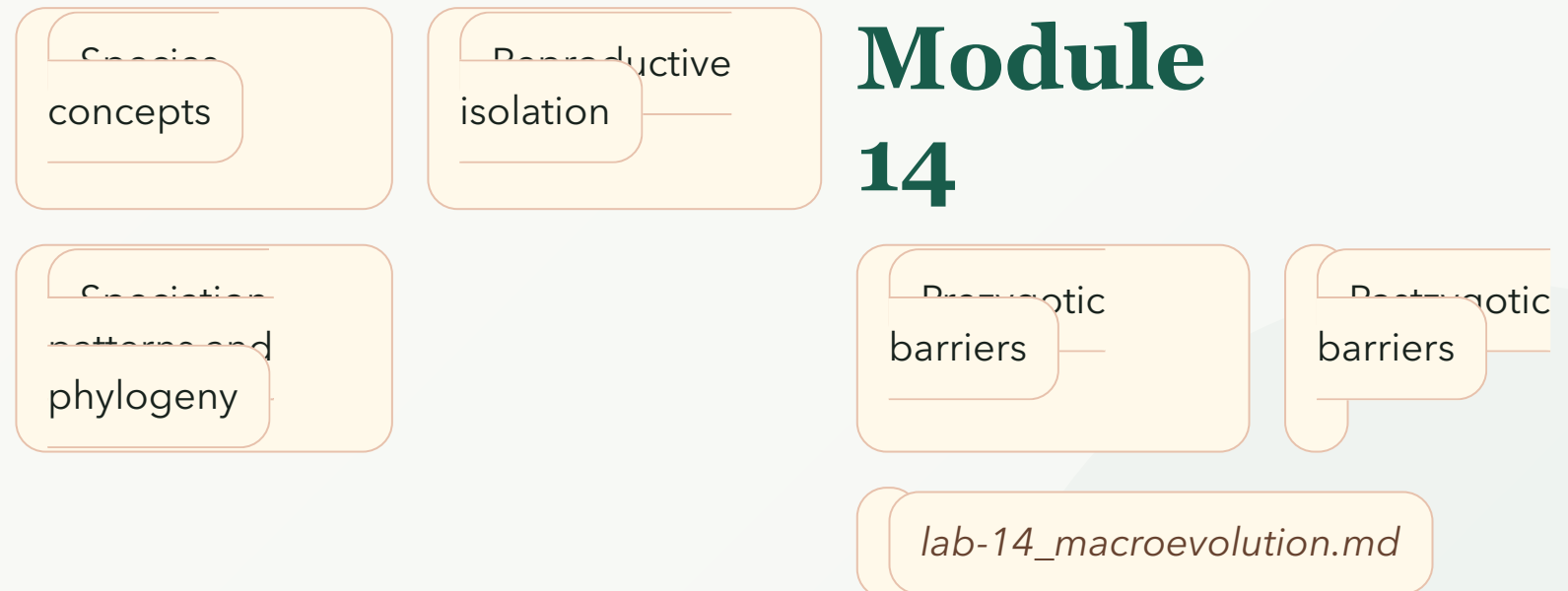


MACROEVOLUTION

Module map

- Module 14: Macroevolution
- Connected lab:
lab-14_macroevolution.md
- Use this deck as the visual path through the module's evidence and retrieval work.



MACROEVOLUTION

Learning objectives

- State the biological species concept and its limits.
- Explain reproductive isolation as reduced gene flow.
- Classify examples as prezygotic or postzygotic barriers.
- Compare allopatric and sympatric speciation.
- Interpret speciation as population-level divergence over time.

OBJECTIVE 1

State the biological species concept and its limits.

OBJECTIVE 2

Explain reproductive isolation as reduced gene flow.

OBJECTIVE 3

Classify examples as prezygotic or postzygotic barriers.

OBJECTIVE 4

Compare allopatric and sympatric speciation.

OBJECTIVE 5

Interpret speciation as population-level divergence over time.

MACROEVOLUTION

Topic sequence

- Species concepts: Species concepts define lineages using reproductive, morphological, or evolutionary criteria
- Reproductive isolation: Reproductive isolation reduces gene flow between populations.
- Prezygotic barriers: Prezygotic barriers prevent mating or fertilization before a zygote forms.
- Postzygotic barriers: Postzygotic barriers reduce hybrid survival or fertility after a zygote forms.
- Speciation patterns and phylogeny: Speciation can occur allopatrically or sympatrically and is represented in phylogenies.

TOPIC 1

Species concepts

TOPIC 2

Reproductive isolation

TOPIC 3

Prezygotic barriers

TOPIC 4

Postzygotic barriers

TOPIC 5

Speciation patterns and phylogeny

MACROEVOLUTION

Module 14: Macroevolution Evidence Map

- Module focus: Macroevolution
- Central claim: Macroevolution explains speciation as reduced gene flow, reproductive isolation, and phylogenetic divergence.
- Key nodes to connect: Species concepts, Reproductive isolation, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-14_macroevolution.md supplies an evidence surface for the map.

Teaching note: Ask students to explain one edge in their own words before moving on.

MACROEVOLUTION

Module 14: Macroevolution Reasoning Sequence

- Module focus: Macroevolution
- Trace the model: Species concepts -> Reproductive isolation -> Prezygotic barriers -> Postzygotic barriers
- Inputs become outputs: Species concepts, Reproductive isolation -> State the biological species concept and its limits., Explain reproductive isolation as reduced gene flow.
- Feedback check:
lab-14_macroevolution.md evidence checks whether students can explain prezygotic barriers.
- Lab connection:
lab-14_macroevolution.md gives students a process to observe or test.

Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

MACROEVOLUTION

Terms as evidence handles

- Species: Lineage or group identified by a species concept.
- Biological species concept: Species defined by interbreeding and reproductive isolation.
- Reproductive isolation: Barriers that reduce gene flow between groups.
- Prezygotic barrier: Barrier before fertilization.
- Postzygotic barrier: Barrier after fertilization.
- Allopatric speciation: Speciation with geographic separation.

SPECIES

Lineage or group identified by a species concept.

BIOLOGICAL SPECIES CONCEPT

Species defined by interbreeding and reproductive isolation.

REPRODUCTIVE ISOLATION

Barriers that reduce gene flow between groups.

PREZYGOTIC BARRIER

Barrier before fertilization.

POSTZYGOTIC BARRIER

Barrier after fertilization.

ALLOPATRIC SPECIATION

Speciation with geographic separation.

MACROEVOLUTION

Lab connection

- Lab file: lab-14_macroevolution.md
- Lab evidence should test or illustrate: Prezygotic barriers
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION

Species concepts

EVIDENCE

lab-14_macroevolution.md

CLAIM

State the biological species concept and its limits.

MACROEVOLUTION

Contrast check

- Do not stop at: Species concepts define lineages using reproductive, morphological, or evolutionary criteria.
- Stronger explanation: Speciation can occur allopatrically or sympatrically and is represented in phylogenies.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Species concepts define lineages using reproductive, morphological, or evolutionary criteria.

DEEPER

Speciation can occur allopatrically or sympatrically and is represented in phylogenies.

MACROEVOLUTION

Module 14: Macroevolution Retrieval and Lab Check

- Module focus: Macroevolution
- Retrieval prompt: What does the biological species concept emphasize?
- Answer check: State the biological species concept and its limits.
- Required terms: Species, Biological species concept, Reproductive isolation
- Lab connection:
lab-14_macroevolution.md;
lab-14_macroevolution.md supplies evidence for macroevolution evidence from isolation and lineage patterns.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

MACROEVOLUTION

Practice quiz bridge

- 1. Which statement best defines Species in Module 14?
- 2. Which learning goal best supports the topic "Reproductive isolation"?
- 3. How should lab-14_macroevolution.md support Module 14?
- 4. Which retrieval move best prepares a student for Macroevolution?

QUIZ 1

A

QUIZ 2

B

QUIZ 3

C

QUIZ 4

D

MACROEVOLUTION

Synthesis and exit ticket

- Exit claim: explain species concepts using evidence from lab-14_macroevolution.md.
- Must include: Species, Biological species concept, Reproductive isolation.
- Revision prompt: what evidence would change your explanation?

CLAIM

Species concepts

EVIDENCE

lab-14_macroevolution.md

REVISION

What would change your mind?